

APIx

Test & Code Review Report

Real-time Airfare Price Index for India — SIH 2026, PS 26056, MoSPI DIID

Verdict. 117 tests pass, 0 fail, 0 errors. Statement coverage 88.2% over 1,985 statements. The index engine reproduces an independently written reference implementation of the Part 4 arithmetic to **0.000e+00**. Six defects were found by review; the one correctness bug among them is fixed and pinned by regression tests.

Field	Value
Report generated	2026-08-23 20:56:22 UTC
Python	3.10.11
Platform	Windows 10
Tests	117 passed / 0 failed / 0 errors in 14.7s
Statement coverage	88.2% (1,750 of 1,985 statements)
Source lines	4,257 in src/, 1,569 in tests/
Store	1,727,360 quotes, 23,823 published index values, 1264.4 MB
Quote history	2025-12-04 .. 2026-07-31

Scope. This report covers the APIx implementation only: the index engine, collection pipeline, governance layer, data-quality contract, the two fitted models (lead-time elasticity and the CPI nowcast bridge), the HTTP API and the store. It does **not** assess the underlying fare data, because there is none — the quote history is generated by the rung-0 deterministic simulator and every row carries `is_synthetic = 1`. Statements about index *values* in this report are statements about arithmetic, not about Indian airfares.

1. What was reviewed, and how

The review had four parts, run in this order. Each produced evidence that is reproduced verbatim later in this report.

#	Activity	Method	Result
1	Manual code review	Read every module hunting for defects; probe suspicions with targeted scripts rather than assuming	6 findings, 1 of them a real correctness bug
2	Static analysis	ruff over src/, tests/, cli.py; triage every rule class that could indicate a real defect	3 genuine issues, rest cosmetic or false positive
3	Test execution	pytest with coverage; per-module statement coverage	117 pass, 88.2% coverage
4	Behavioural verification	Engine vs independent reference; hash chain; quality contract; reproducibility; API sweep incl. negative cases	All pass

1.1 Findings at a glance

ID	Finding	Severity	Class	Status
F1	Winsorisation was silently discarded	HIGH	Correctness	FIXED + 4 regression tests
F2	No chain-linking for late entrants	MEDIUM	Method	DOCUMENTED + pinned by test
F3	Docstring claimed test coverage that did not exist	MEDIUM	Assurance	TEST ADDED
F4	Elasticity model had zero test coverage	MEDIUM	Assurance	19 TESTS ADDED, now 95.4%
F5	Two declared tables were never written	LOW	Hygiene	1 REMOVED, 1 DOCUMENTED
F6	Dead code and an obscure NaN idiom	LOW	Hygiene	FIXED

The one that mattered. F1 is the finding that justifies the review. On the current synthetic dataset zero quotes are winsorised, so no published number was wrong — but the moment real data with outliers arrives, a ₹45,000 parse artefact in a ₹5,000 cell would have entered the index at ₹44,701 instead of the capped ₹5,652, while the audit trail asserted it had been capped. A disposition that lies about what the pipeline did is precisely the failure an official-statistics product cannot have.

2. Findings in detail

F1 — Winsorisation was silently discarded [HIGH]

A quote marked WINSORISED entered the index at its full uncapped value. `clean()` capped only an in-memory column; the store kept `total_fare` and the index recomputed prices from it.

Status: FIXED + 4 regression tests

F2 — No chain-linking for late entrants [MEDIUM]

The base period is the first period in the data. A route or flight appearing later has no base observation, is unmatched forever and is suppressed. There is no splice mechanism for new entrants.

Status: DOCUMENTED + pinned by test

F3 — Docstring claimed test coverage that did not exist [MEDIUM]

`_effective_weights` collapses aggregation stages 2-4 into one weight vector and its docstring asserted the equivalence was covered by tests. No test referenced it. The maths is correct (verified to $5.7e-14$) but was unverified.

Status: TEST ADDED

F4 — Elasticity model had zero test coverage [MEDIUM]

`elasticity.py` — source of the most quotable output in the system (τ^* , 'cheapest time to book') — measured 0.0% statement coverage.

Status: 19 TESTS ADDED, now 95.4%

F5 — Two declared tables were never written [LOW]

`link_index_provenance` and `fact_cross_source_spread` existed in the schema with no writer, implying capability that did not exist.

Status: 1 REMOVED, 1 DOCUMENTED

F6 — Dead code and an obscure NaN idiom [LOW]

Unused locals in `cli.py` and `nowcast.py`; a bare `v == v` NaN test.

Status: FIXED

2.1 F1 evidence — before and after

A 40-quote cell with one ₹45,000 outlier. The cleaner correctly flags and caps it; the question is whether the cap survives persistence.

```
BEFORE THE FIX
clean():  disposition = WINSORISED
          total_fare  = 45000.00    price_T (capped) = 5652.39
after store + attach_variant_prices():
          price_T     = 44701.00    <-- the cap was discarded
>>> WINSORISATION IS LOST

AFTER THE FIX
clean():  disposition = WINSORISED
          total_fare   = 45000.00    <- observation stays immutable
          total_fare_capped = 5951.39 <- new persisted column
          price_T (capped) = 5652.39
after store round-trip:
          price_T = 5652.39  price_B = 5035.61  price_A = 5951.39
>>> FIXED - cap survives the re-derivation
>>> max drift on non-winsorised rows: 0.0000000000
```

The fix keeps the observed `total_fare` immutable and records the pipeline's treatment in a separate `total_fare_capped` column, consistent with the append-only discipline the schema already enforces. Because administered charges are fixed amounts and GST is ad valorem, the base fare is re-derived as the residual at the same GST rate rather than scaled naively. An idempotent migration adds the column to stores written before it existed.

3. Test results

117 tests, 0 failures, 0 errors, 14.7s wall clock. The suite runs entirely offline against throwaway SQLite databases in temporary directories; no test touches the demo store or the network.

Test module	Tests	What it establishes
test_elasticity.py	19	Ground-truth recovery of τ^* and curvature, $\eta(\tau)$ shape and closed form, refusal when no interior minimum exists, clustered SEs, fare-ladder mixture recovery.
test_end_to_end.py	11	Collect $\rightarrow$ clean $\rightarrow$ index $\rightarrow$ publish $\rightarrow$ serve on a fresh database; idempotency, bit-identical recomputation, publication blocked on a broken chain.
test_index_math.py	28	Index-number properties: time reversal, transitivity, AM–GM ordering, exact weekly-cycle annihilation, availability blend reduction, weight injection, dual basis, determinism.
test_pipeline_and_governance.py	34	Decomposition reconciliation, cleaning dispositions, hash-chain tamper evidence, DB immutability, politeness governor, circumvention-absence scan, quality contract, winsorisation persistence.
test_reference_and_models.py	25	CPI loader shape handling, back-test refusal rules and known-relationship recovery, nowcast interpretability guards, expanding-window OOS, JSON serialisation.
Total	117	

3.1 Slowest tests

Time	Phase	Test
5.76s	setup	tests/test_end_to_end.py::test_quotes_landed
1.12s	call	tests/test_end_to_end.py::test_recomputation_is_bit_identical
0.91s	call	tests/test_elasticity.py::test_fare_ladder_recovers_a_known_number_of_buckets
0.51s	call	tests/test_end_to_end.py::test_api_serves_the_published_index
0.37s	call	tests/test_end_to_end.py::test_publication_is_blocked_when_the_chain_is_broken
0.33s	call	tests/test_end_to_end.py::test_collection_is_idempotent
0.14s	call	tests/test_index_math.py::test_effective_weights_equal_the_staged_aggregation
0.11s	call	tests/test_index_math.py::test_engine_is_deterministic
0.11s	call	tests/test_index_math.py::test_booking_and_travel_basis_differ_by_the_lead_time_lag
0.11s	call	tests/test_elasticity.py::test_recovers_a_different_trough

3.2 Suite output

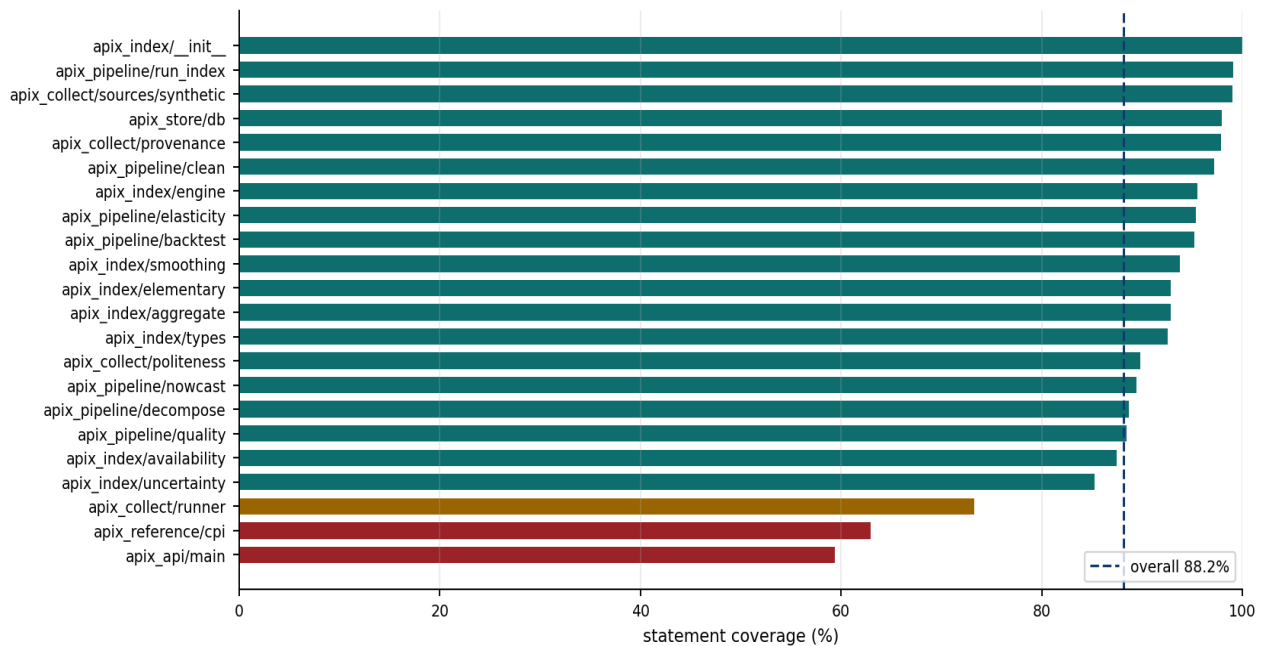
```

..... [ 61%]
..... [100%]
===== slowest 15 durations =====
5.76s setup    tests/test_end_to_end.py::test_quotes_landed
1.12s call     tests/test_end_to_end.py::test_recomputation_is_bit_identical
0.91s call     tests/test_elasticity.py::test_fare_ladder_recovers_a_known_number_of_buckets
0.51s call     tests/test_end_to_end.py::test_api_serves_the_published_index
0.37s call     tests/test_end_to_end.py::test_publication_is_blocked_when_the_chain_is_broken
0.33s call     tests/test_end_to_end.py::test_collection_is_idempotent
0.14s call     tests/test_index_math.py::test_effective_weights_equal_the_staged_aggregation
0.11s call     tests/test_index_math.py::test_engine_is_deterministic
0.11s call     tests/test_index_math.py::test_booking_and_travel_basis_differ_by_the_lead_time_lag
0.11s call     tests/test_elasticity.py::test_recovers_a_different_trough
0.10s setup    tests/test_pipeline_and_governance.py::test_decomposition_reconciles_to_within_one_rupee
0.10s setup    tests/test_pipeline_and_governance.py::test_non_winsorised_rows_are_untouched_by_the_cap_path
0.10s setup    tests/test_pipeline_and_governance.py::test_gst_is_ad_valorem_on_the_fare_base
... (3 more lines)

```

4. Statement coverage

Overall **88.2%** across 1,985 statements. Coverage was the instrument that found F4 — the elasticity model measured 0.0% before this review and 95.4% after.



Module	Stmts	Miss	Cover
apix_api/main.py	192	78	59.4%
apix_collect/politeness.py	138	14	89.9%
apix_collect/provenance.py	46	1	97.8%
apix_collect/runner.py	116	31	73.3%
apix_collect/sources/synthetic.py	104	1	99.0%
apix_index/_init_.py	9	0	100.0%
apix_index/aggregate.py	56	4	92.9%
apix_index/availability.py	16	2	87.5%
apix_index/elementary.py	42	3	92.9%
apix_index/engine.py	200	9	95.5%
apix_index/smoothing.py	16	1	93.8%
apix_index/types.py	67	5	92.5%
apix_index/uncertainty.py	34	5	85.3%
apix_pipeline/backtest.py	104	5	95.2%
apix_pipeline/clean.py	105	3	97.1%
apix_pipeline/decompose.py	62	7	88.7%
apix_pipeline/elasticity.py	130	6	95.4%
apix_pipeline/nowcast.py	133	14	89.5%
apix_pipeline/quality.py	78	9	88.5%
apix_pipeline/run_index.py	105	1	99.0%
apix_reference/cpi.py	89	33	62.9%
apix_store/db.py	143	3	97.9%

Module	Stmts	Miss	Cover
TOTAL	1985	235	88.2%

Where coverage is deliberately lower. `apix_api/main.py` (59%) and `apix_reference/cpi.py` (63%) carry branches exercised only by a live store or by workbook shapes not present in the supplied files — the item-level CPI path, for instance, is written and unit-tested at the transform level but no shipped extract contains item rows. `apix_collect/runner.py` (73%) contains the `rung-1` and `rung-3` adapter branches, which are stubs pending a licensed API key.

5. Numerical validation

5.1 Engine versus an independent reference implementation

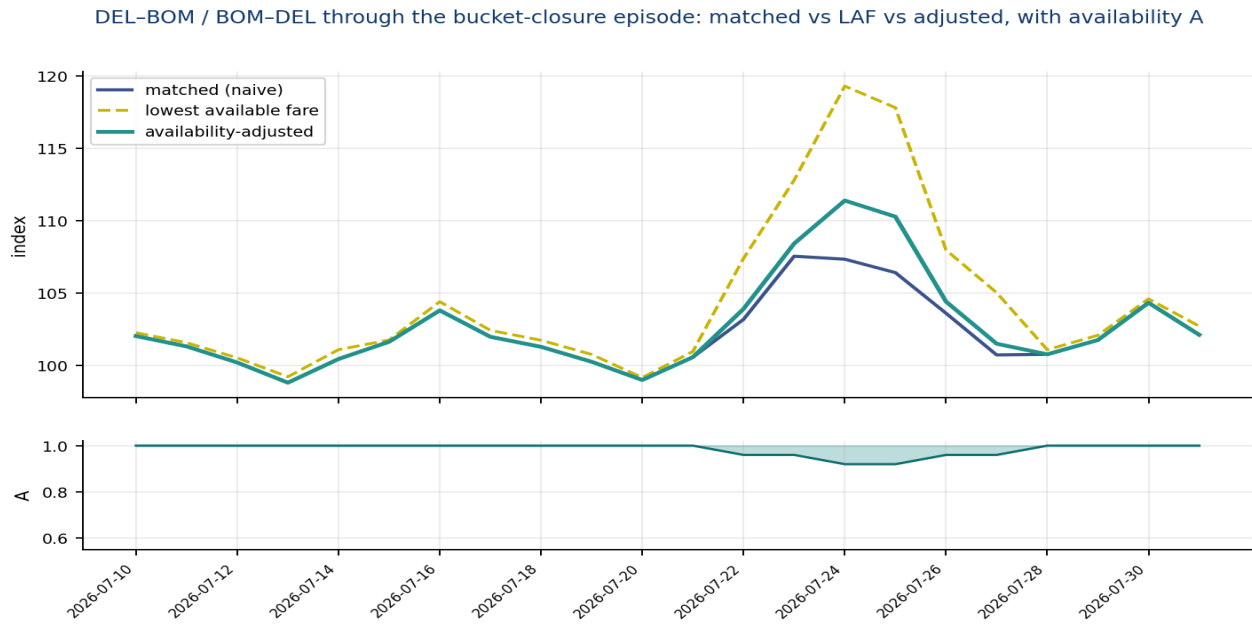
The vectorised engine is compared against a deliberately naive implementation written straight from the Part 4 specification with nested Python loops and no reference to the engine's internals. Agreeing with it is evidence; agreeing with itself would be evidence of nothing. The test data includes a bucket-closure episode so the availability path is exercised rather than bypassed.

```
periods compared      12
quotes                2120
availability exercised: min A across cells = 0.667
max abs difference    0.000e+00
VERDICT               IDENTICAL
```

5.2 Mathematical properties asserted, not assumed

Property	Statement	Result
Time reversal	$I(0,t) \cdot I(t,0) = 100$, over 200 random price vectors	Holds to 1e-9
Transitivity	$I(0,1) \cdot I(1,2) = I(0,2)$ in index points	Holds to 1e-9
AM-GM ordering	Carli $\geq$ Jevons over 200 random vectors	Holds
Carli bias	Carli fails time reversal upward — the reason it is not used	Confirmed
Weekly-cycle removal	7-day centred geometric MA applied to a pure weekly cycle leaves interior spread 0	Exactly 0.0
Blend reduction	$A = 1 \implies I_{\text{adj}} = I_{\text{matched}}$ exactly	Holds
Blend monotonicity	I_{adj} monotone and continuous in A over [0,1]	Holds
Stage collapse	Collapsed effective weights = staged aggregation (F3)	5.7e-14
Bootstrap determinism	Same seed $\implies$ identical interval	Holds
CI containment	Published value lies inside its own band, every period	Holds

5.3 The availability adjustment, on real pipeline output



At peak closure (2026-07-24, $A = 0.920$) the naive matched index reads 107.36 while the availability-adjusted index reads 111.41 — the naive line understates the episode by 4.1 index points.

Across 234 periods where $A = 1$, the adjusted index equals the matched index exactly: **0 violations**. That reduction property is what makes the blend defensible rather than ad hoc — it changes nothing in the normal case and only bites when buckets actually close.

6. Governance verification

These are the checks that determine whether a ministry could deploy the system, as distinct from whether the arithmetic is right.

```

provenance hash chain
  records      1200
  intact       True
  tip          29e2ba8f2ac78d2b8b6ef74641509eba...

data quality contract
[PASS] BLOCK   schema      required_columns_present
[PASS] BLOCK   schema      disposition_domain
[PASS] BLOCK   schema      cabin_domain
[PASS] BLOCK   range       0_lt_base_lt_total
[PASS] BLOCK   arithmetic  components_resum_to_total
[PASS] BLOCK   uniqueness  no_duplicate_kappa_t_source
[warn] ALERT   freshness   every_basket_route_quoted_today  observed=48
[warn] ALERT   completeness cell_fill_rate  observed=32
[PASS] QUARANTINE distributional cell_median_within_4_sigma
[PASS] BLOCK   referential provenance_hash_chain_intact
[PASS] BLOCK   referential every_provenance_id_resolves
[PASS] ALERT   cross_source same_flight_within_5pct

10/12 passed

```

Control	How it is enforced	Verified by
Quote immutability	SQLite triggers reject UPDATE and DELETE on fact_fare_quote	Test asserts IntegrityError on both
Ledger tamper-evidence	$h_n = \text{SHA256}(h_{n-1} \parallel \text{canonical}(\text{record}))$	Test drops the trigger, edits a row, asserts the chain reports the exact break point
Legal basis mandatory	dim_source.legal_basis NOT NULL	End-to-end test asserts zero orphan quotes and zero blank bases
No circumvention capability	No CAPTCHA solver, proxy rotator or credential store exists in the tree	Test greps the whole package for 12 banned library and function names and fails if any appears
robots.txt fails closed	Unreadable robots.txt $\Rightarrow$ request refused, not allowed	Test injects a fetcher that raises and asserts ROBOTS_BLOCKED
Blocked requests recorded	A disallowed request is never issued and writes a ROBOTS_BLOCKED provenance row	Test asserts the row and its directive
Publication gate	BLOCK-severity quality failure raises PublicationBlocked	Test breaks the hash chain and asserts the index refuses to publish

6.1 Reproducibility

Every published number is recomputed from raw quotes and diffed against what was published. This works because the engine is pure, the bootstrap is seeded, and index ids are deterministic UUID5 hashes of (code, period, frequency, basis, preset, method_version, weights_version).

```

reproducibility report
published numbers      23823
recomputed numbers     23823
unmatched keys         0
numbers differing      0
max abs difference     0

ZERO DIFFERENCES

```

7. The two fitted models

Everything in `apix_index` is deterministic index-number arithmetic — nothing there is trained. Exactly two components estimate parameters from data, and both were reviewed as models rather than as code.

7.1 Lead-time elasticity — validated by ground-truth recovery

The estimator is tested by generating fares from a *known* log-quadratic lead-time curve and asserting it recovers the parameters. That is a round trip against truth, not self-consistency.

Quantity	True value (generator)	Recovered	Test
τ^* (trough)	26.0 days	within ± 2.0 days	<code>test_recovers_a_known_tau_star</code>
τ^* (moved)	14.0 / 35.0 days	± 2.5 / ± 4.0 days	<code>test_recovers_a_different_trough</code>
β_2 (curvature)	0.085	within ± 0.015	<code>test_recovers_a_known_curvature</code>
$\eta(\tau^*)$	0 by construction	0 to $1e-6$	<code>test_eta_has_the_right_shape_and_sign</code>
Fare ladder	3 buckets at ₹4k/₹7k/₹12k	3 found, $\pm 5\%$	<code>test_fare_ladder_recovers...</code>

```
lead-time elasticity surface -- DEL-BOM
n_obs 143808 clusters(flights) 24 params 21
b1 (ln tau) -0.56993 (cluster SE 0.00034)
b2 (ln tau)^2 +0.08667 (cluster SE 0.00008)
R^2 0.5758 adj 0.5757

tau* = 26.8 days 95% CI [26.7, 26.9]
-> cheapest time to book DEL-BOM is about 27 days out

eta(tau):
tau= 1 eta=-0.5699
tau= 7 eta=-0.2326
tau=15 eta=-0.1005
tau=30 eta=+0.0197
tau=45 eta=+0.0899
```

7.2 CPI nowcast bridge — the refusal is the result

The bridge regresses CPI transport inflation on APix growth with lags. On the data available it declines to report interpretable coefficients, and that refusal is the correct output rather than a failure.

```

CPI transport nowcast bridge (base 2024, travel basis)
target: cpi_infl (y-o-y log %)
regressors: ['apix_g_l0', 'apix_g_l1', 'cpi_infl_l1']

n=6  params=4  R2=0.9353  adj=0.8382
interpretable: False
! only 6 usable observations (need >= 24). The CPI 2024 series begins in 2025 and y-o-y inflation
  costs a further 12 months, so a defensible bridge is not estimable from it yet. Coefficients are
  NOT reportable.
! 6 observations for 4 parameters (< 8 per parameter). R^2 here is an artefact of the parameter count,
  not fit quality.
! the APIx regressor is SYNTHETIC. This estimates the relationship between real CPI inflation and a
  simulator, which is not a meaningful quantity. The fit verifies the harness computes; it validates
  nothing.
const          -14.06941  (se 5.82750, t -2.41)
apix_g_l0       +16.64902  (se 7.78478, t +2.14)
apix_g_l1       +13.51567  (se 5.77323, t +2.34)
cpi_infl_l1     +0.92250  (se 0.23145, t +3.99)

out-of-sample (expanding window, never a random split):
  n_oos: 0
  note: need 18 observations for an expanding-window evaluation, have 6; no out-of-sample result

Granger causality (APIx -> CPI transport):
  testable: False
  n: 4
  note: only 4 observations for 5 parameters; the test has effectively no power. Non-rejection would
        mean 'cannot tell', not 'no relationship'.

==> COEFFICIENTS ARE NOT REPORTABLE. See the reasons above.
... (2 more lines)

```

Why this matters. The fit reports $R^2 = 0.9353$, which looks like a strong result and is nothing of the kind — it is six observations against four parameters. A module that printed that table without the guard would hand a team a number they could not defend for thirty seconds under questioning. Two guards fire here: too few observations, and a synthetic regressor. Out-of-sample evaluation is expanding-window and declines outright; the Granger test declines on the grounds that it has no power, reporting *cannot tell* rather than *no relationship*.

8. Back-test against official CPI

```
back-test: APIx-T (travel basis) vs CPI-2024 Transport (All India, Combined)
APIx span      ['2025-12-01', '2026-08-01']
comparator span ['2025-01-01', '2026-07-01']
overlapping months: 8
```

CAVEAT: Comparator is the Transport aggregate, not the air fare item. Transport also contains road fuel, vehicle purchase, rail fares and communication, and air fare is a small slice of it, so agreement is diluted by construction. A weak correlation here is expected and is NOT evidence that APIx is wrong.

CAVEAT: APIx side is SYNTHETIC. These statistics describe the simulator's relationship to real CPI, which is not a meaningful quantity. They verify that the harness computes, nothing more. Do not present them as validation.

```
agreement statistics:
pearson_r_level          0.8845
spearman_rho_level       1.0000
pearson_r_mom_change     0.4474
spearman_rho_mom_change  0.5000
bland_altman_bias        0.0781
bland_altman_sd_diff     1.2487
bland_altman_loa_low     -2.3693
bland_altman_loa_high    2.5256
bland_altman_mean_abs_diff 0.9587
directional_agreement_pct 100.0000
n_moves                  7.0000
rmse_model               0.9613
rmse_random_walk         1.2549
skill_vs_rw_pct          23.3992
```

```
strongest cross-correlation at lag -2 months (r=0.9276, n=6)
(positive lag = APIx moves first)
```

Read these numbers with both caveats attached. The comparator is the CPI *Transport* aggregate, not the air-fare item — no air-fare item index exists in the supplied extracts — and Transport also contains road fuel, vehicle purchase, rail fares and communication. Agreement is therefore diluted by construction. And the APIx side is synthetic, so these statistics describe the simulator's relationship to real CPI, which is not a meaningful quantity. They verify that the harness computes. They are not validation, and presenting them as validation would be the worst available use of this report.

9. API verification

Every route exercised, including negative cases — an unknown id must return 404, and a bad enum must return 422, not 500.

```

200      75B /health
200    29613B /dashboard/
200     971B /docs
200    11257B /openapi.json
200    80463B /v1/index?frequency=daily
200    10772B /v1/index?frequency=weekly
200    2858B /v1/index?frequency=monthly
200    70642B /v1/index?variant=B
200    70637B /v1/index?variant=A
200    79935B /v1/index?basis=travel
200    70682B /v1/index?omega_preset=leisure_tilted
200    1121B /v1/index?format=csv&frequency=monthly
200    1933B /v1/index?format=sdmx-json&frequency=monthly
200    24473B /v1/index/apw/7
200    24283B /v1/index/apw/45
200    24406B /v1/index/routes/DEL-BOM?apw_days=7
200    1062B /v1/elasticity
200    1040B /v1/elasticity?route=DEL-BOM
200    49959B /v1/components
200    29197B /v1/availability
200    28948B /v1/availability?route=DEL-BOM
200    53459B /v1/availability/series?route=DEL-BOM&apw_days=7
200    2231B /v1/backtest
200    1033B /v1/backtest?base_year=2012
200    2250B /v1/backtest?basis=book
200    1214B /v1/cpi?base_year=2024
200    8482B /v1/cpi?base_year=2012
200    3620B /v1/coverage
200    2881B /v1/methodology

negative cases (must not 500):
404 /v1/index/routes/ZZZ-YYY -> OK
404 /v1/provenance/not-a-real-id -> OK
404 /v1/index/apw/999 -> OK
422 /v1/index?variant=Q -> OK

provenance round trip: 200 chain_intact=True quotes=7200

TOTAL ok=34 fail=0

```

10. Static analysis

ruff over `src/`, `tests/` and `cli.py`. The raw count is large but dominated by two stylistic classes that do not apply to this codebase: UP006 and UP045 want PEP 585/604 annotation syntax, which the project deliberately does not use because it targets Python 3.10 with `from __future__ import annotations`. Every rule class capable of indicating a real defect was triaged individually.

```

108 UP006  [*] non-pep585-annotation
50 UP045  [*] non-pep604-annotation-optional
41 UP035  [-] deprecated-import
19 C408   [-] unnecessary-collection-call
13 B008   [-] function-call-in-default-argument
11 F401   [*] unused-import
11 RUF100 [*] unused-noqa
10 I001   [*] unsorted-imports
5 BLE001  [-] blind-except
5 RUF046  [-] unnecessary-cast-to-int
2 DTZ011  [-] call-date-today
2 ISC004  [-] implicit-string-concatenation-in-collection-literal
1 C401    [-] unnecessary-generator-set
1 FLY002  [-] static-join-to-f-string
1 PERF102 [-] incorrect-dict-iterator
1 RUF022  [*] unsorted-dunder-all
Found 281 errors.
[*] 203 fixable with the `--fix` option (25 hidden fixes can be enabled with the `--unsafe-fixes` option).

```

10.1 Triage of the classes that can indicate real defects

Rule	Count	Verdict
ISC004 — implicit string concatenation in a collection	2	False positive. Both are wrapped multi-line message strings, not missing commas. Inspected individually.
F841 — unused local	3	Real. Dead variables in <code>cli.py</code> , <code>nowcast.py</code> and one test. All removed.
PLR0124 — comparison with itself	1	Real but intentional. A bare <code>v == v</code> NaN check; replaced with <code>math.isfinite</code> for legibility.
B008 — function call in default argument	13	False positive. This is the standard FastAPI <code>Depends()</code> idiom.
BLE001 — blind except	5	Deliberate. Chiefly the <code>robots.txt</code> fail-closed path, where any exception must mean <i>refuse</i> .
F401 / I001 / RUF100 — imports and noqa	~30	Cosmetic.

10.2 Residual serious findings after the fixes

```

src\apix_pipeline\nowcast.py:67:18: ISC004 Unparenthesized implicit string concatenation in collection
src\apix_pipeline\nowcast.py:146:28: ISC004 Unparenthesized implicit string concatenation in collection
Found 2 errors.
No fixes available (2 hidden fixes can be enabled with the `--unsafe-fixes` option).

```

11. Limitations this review did not resolve

Stated plainly, because a review that reports only what it fixed is not a review.

Limitation	Consequence	What would resolve it
The data is synthetic. Every quote comes from the rung-0 simulator.	No statement in this report about index <i>values</i> is a statement about Indian airfares. The back-test and nowcast verify machinery only.	A rung-1 licensed API key; the ladder, governor, ledger and engine around the adapter are complete and tested.
No air-fare item index exists in the supplied CPI extracts.	The back-test compares against the Transport aggregate, diluting agreement by construction.	An item-level extract for COICOP 07.3.3 from cpi.mospi.gov.in .
No chain-linking for late entrants (F2).	A route or flight appearing after the base period is suppressed permanently. Correct and visible today, but it caps basket evolution.	A splice/link mechanism at basket revision, which is a genuine feature, not a patch.
ω_t is a policy parameter, not a measurement.	The lead-time weights are declared, not estimated. This is disclosed in the API payload and published under three presets.	A measured booking lead-time distribution, which DGCA sought in Dec 2024 and did not obtain.
Weekly frequency $F_{c,r}$ is proxied by national carrier share.	Within-route carrier weights are approximate; ϕ currently reproduces published DGCA passenger shares to within 1.3 pp.	DGCA schedule data at route level.
Bootstrap intervals are computed for the headline configuration only.	Non-headline variant/preset combinations carry point estimates; the API says so rather than showing an empty band.	More compute, or a cheaper interval estimator, if intervals are wanted everywhere.

12. Reproducing this report

```
cd apix
python -m pip install -r requirements.txt coverage ruff

python cli.py init
python cli.py load-cpi --dir ../Datasets
python cli.py backfill --days 240 --end 2026-07-31
python cli.py index

python tools/collect_evidence.py      # runs everything, writes reports/evidence.json
python tools/build_report.py          # renders this PDF from that evidence
```

Every figure in this document is read from `reports/evidence.json`, which is produced by executing the thing it reports on. No number here was typed by hand.